

# Fair, Interpretable, and Actionable Readmission Risk Modeling

Diabetes 130-US Hospitals (1999–2008)  
Project 02 | Digital Health & Medical Data Science Track

## 1. Project Goal

Build an educational risk-stratification model that predicts 30-day early hospital readmission (<30) among diabetic inpatients, with emphasis on interpretability, honest evaluation, and fairness across patient subgroups.

This is not a deployed clinical decision system.

## 2. Dataset

- Source: UCI Machine Learning Repository — Diabetes 130-US Hospitals for Years 1999–2008
- Size: 101,766 encounters, 50 attributes, 130 US hospitals
- Target: binary early readmission (<30 vs. not), approximately 11.2% positive class
- DOI: 10.24432/C5230J | License: CC BY 4.0

## 3. Data Preparation

### 3.1 Dropped Columns

- Identifiers: encounter\_id, patient\_nbr
- High missingness: weight, max\_glu\_serum, A1Cresult, medical\_specialty, payer\_code
- Raw diagnosis codes: diag\_1, diag\_2, diag\_3 (replaced by grouped categories)
- Target and near-constant columns: readmitted, examide, citoglipton

### 3.2 ICD-9 Grouping

Diagnosis codes were mapped to clinical categories: Circulatory, Respiratory, Digestive, Diabetes, Injury, Musculoskeletal, Genitourinary, Neoplasms, and Other.

### 3.3 Feature Engineering

- $\text{total\_prior\_visits} = \text{number\_outpatient} + \text{number\_emergency} + \text{number\_inpatient}$
- $\text{meds\_per\_day} = \text{num\_medications} / (\text{time\_in\_hospital} + 1)$
- $\text{lab\_per\_day} = \text{num\_lab\_procedures} / (\text{time\_in\_hospital} + 1)$

Rationale: capture overall prior utilization intensity and treatment/lab intensity relative to length of stay.

### 3.4 Encoding and Split

Categorical variables were label-encoded. Data were split 80/20 with stratification and random\_state=42. Class weight was set as scale\_pos\_weight  $\approx 5.57$  (full ratio  $\times 0.7$ ).

## 4. Models

### 4.1 XGBoost and LightGBM

Final models aligned with strong 2024–2025 papers on this dataset.

- `n_estimators=300`, `max_depth=6`, `learning_rate=0.05`
- `subsample=0.8`, `colsample_bytree=0.8`
- `scale_pos_weight` for class imbalance

XGBoost grows trees level-wise; LightGBM grows leaf-wise and is typically faster.

## 4.2 Threshold Tuning

Thresholds from 0.20 to 0.60 were evaluated on a validation split. The threshold maximizing F1 for the <30 class was selected.

## 5. Results

**LightGBM: AUC  $\approx$  0.684 | F1  $\approx$  0.288 | Recall  $\approx$  0.47**

**XGBoost: AUC  $\approx$  0.682 | F1  $\approx$  0.287 | Recall  $\approx$  0.45**

These values match honest recent literature (typical AUC 0.66–0.70). Accuracy alone is misleading due to class imbalance.

## 6. Feature Importance (Q7)

Top features from XGBoost:

- 1. `number_inpatient`
- 2. `discharge_disposition_id`
- 3. `total_prior_visits`
- 4. `diabetesMed`
- 5. `diag1_cat`
- 6. `number_diagnoses`
- ... (`meds_per_day` also ranked in the top 12)

Figure: `figures/feature_importance_xgb.png`

## 7. Fairness (Q8)

Model recall was compared across race, gender, and age brackets. Differences exist and were reported transparently. Figure: `figures/08_fairness_recall.png`

## 8. Limitations and Recommendation (Q9)

- Data period 1999–2008; ICD-9 only; high missingness in key variables
- Encounter-level split (not purely patient-level)
- No external validation on contemporary data

Recommendation: use only as a risk-stratification aid for discharge planning — never for automated care denial or approval.

## 9. Mapping to Project Questions

- Q1–Q3: Population description, group comparisons, associations (EDA and preprocessing)
- Q4: Baseline and strong models (XGBoost / LightGBM)
- Q5–Q6: Class-wise evaluation and error analysis
- Q7: Feature importance
- Q8: Fairness across subgroups
- Q9: Clinical interpretation and limitations

## 10. How to Run

```
pip install -r requirements.txt
```

```
python src/final_best_pipeline.py
```

```
jupyter notebook notebooks/analysis.ipynb
```

## 11. Repository Structure

data/ | figures/ | notebooks/analysis.ipynb | src/final\_best\_pipeline.py | report/ | README.md | requirements.txt